CSDF-4: Log capturing and event correlation

Squid and Sarg commands (run on Ubuntu)

```
sudo apt update && sudo apt -y install squid sarg
sudo nano /etc/squid/squid.conf      # http_port, acl, http_access
sudo systemctl restart squid
sudo nano /etc/sarg/sarg.conf        # access_log /var/log/squid/access.log
                                     # output_dir /var/www/html/squid-reports
                                     # date_format e ; overwrite_report yes
sudo sarg -x                         # generate the report
# open http://localhost/squid-reports in a browser
```

Sample Squid log (access.log)

```
1761640800.120    245 192.168.60.11 TCP_MISS/200 15230 GET http://www.example.com/index.html user1
    DIRECT/93.184.216.34 text/html
1761640803.455     12 192.168.60.11 TCP_HIT/200 15230 GET http://www.example.com/index.html user1 NONE/-
     text/html
1761640810.002    310 192.168.60.12 TCP_MISS/200 482100 GET http://files.testsite.local/lab.zip user2
    DIRECT/10.0.0.5 application/zip
1761640815.777      3 192.168.60.13 TCP_DENIED/403 3920 GET http://blocked.example.org/ user3 NONE/-
    text/html
1761640816.100      2 192.168.60.13 TCP_DENIED/403 3920 GET http://blocked.example.org/ user3 NONE/-
    text/html
1761640817.350      2 192.168.60.13 TCP_DENIED/403 3920 GET http://blocked.example.org/ user3 NONE/-
    text/html
1761640820.901    120 192.168.60.12 TCP_MISS/200 9800 GET http://www.example.com/style.css user2
    DIRECT/93.184.216.34 text/css
1761640825.640      8 192.168.60.11 TCP_HIT/200 9800 GET http://www.example.com/style.css user1 NONE/-
    text/css
1761640830.210    150 192.168.60.14 TCP_MISS/407 1700 GET http://www.example.com/ - NONE/- text/html
1761640831.015    148 192.168.60.14 TCP_MISS/407 1700 GET http://www.example.com/ - NONE/- text/html
1761640832.330    151 192.168.60.14 TCP_MISS/407 1700 GET http://www.example.com/ - NONE/- text/html
1761640840.470    290 192.168.60.13 TCP_MISS/200 75000 GET http://news.testsite.local/ user3
    DIRECT/10.0.0.5 text/html
```

Program (log_correlate.py)

```
from collections import defaultdict, Counter
from datetime import datetime

users, sites = defaultdict(lambda: [0, 0, 0]), Counter()   # per user: [requests, bytes, hits]
denied, auth_fail, downloads = defaultdict(Counter), Counter(), []

for line in open("access.log"):
    ts, el, ip, code, size, method, url, user, hier, mime = line.split()
    status, size = code.split("/")[1], int(size)
    host = url.split("/")[2]
    u = users[ip]; u[0] += 1; u[1] += size
    if "HIT" in code: u[2] += 1
    sites[host] += 1
    if "DENIED" in code:  denied[ip][host] += 1
    if status == "407":   auth_fail[ip] += 1
    if size > 100000:     downloads.append((datetime.fromtimestamp(float(ts)).strftime("%d/%m/%Y
        %H:%M:%S"), ip, url, size))

print("=== SUMMARY PER CLIENT ===")
print(f"{'IP':15} {'REQ':>4} {'BYTES':>8} {'CACHE-HIT%':>10}")
for ip, (r, b, h) in sorted(users.items(), key=lambda x: -x[1][1]):
    print(f"{ip:15} {r:4d} {b:8d} {100*h/r:9.0f}%")
print("\n=== TOP SITES ===")
for s, c in sites.most_common(3): print(f"{s:25} {c} requests")
print("\n=== DOWNLOADS (> 100 KB) ===")
for d in downloads: print(*d)
print("\n=== EVENT CORRELATION ALERTS ===")
for ip, c in denied.items():
    for s, n in c.items():
        if n >= 3: print(f"[ALERT] {ip} repeatedly denied for {s} ({n} times in 2 s) -> possible policy
            violation")
for ip, n in auth_fail.items():
    if n >= 3: print(f"[ALERT] {ip} had {n} authentication failures (HTTP 407) -> possible brute force /
         bad credentials")
```

Output

```
=== SUMMARY PER CLIENT ===
IP               REQ    BYTES CACHE-HIT%
192.168.60.12      2   491900         0%
192.168.60.13      4    86760         0%
192.168.60.11      3    40260        67%
192.168.60.14      3     5100         0%

=== TOP SITES ===
www.example.com           7 requests
blocked.example.org       3 requests
files.testsite.local      1 requests

=== DOWNLOADS (> 100 KB) ===
28/10/2025 08:40:10 192.168.60.12 http://files.testsite.local/lab.zip 482100

=== EVENT CORRELATION ALERTS ===
[ALERT] 192.168.60.13 repeatedly denied for blocked.example.org (3 times in 2 s) -> possible policy
    violation
[ALERT] 192.168.60.14 had 3 authentication failures (HTTP 407) -> possible brute force / bad credentials
```